

Euclid's Elements — Book I

Proposition 5

Formal Reconstruction — Technical Documentation

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Base angles of an isosceles triangle

1 Introduction

Proposition I.5 is Euclid's first systematic theorem about isosceles triangles. From

$$AB \cong AC$$

it proves both the equality of the interior base angles

$$\angle ABC \cong \angle ACB$$

and, after the equal sides are produced beyond B and C , the equality of the angles below the base,

$$\angle CBD \cong \angle BCE.$$

The classical proof is the celebrated *pons asinorum*. It is not a one-line application of SAS. Euclid extends the two equal sides, introduces two further points F, G , joins them crosswise to the opposite base vertices, and applies Proposition I.4 twice. Common Notion 3 is used first on segments and then on angles.

The current formal reconstruction is organized differently. At the public Book I level, Proposition I.5 merely composes two established theorems: `hilbert_isosceles_base_angles` and `hilbert_adjacent_angles_congruent`. The present audit descends one layer below both calls. The first is a short Pappus-style self-comparison by SAS; the second is Hilbert's Theorem 14 and contains a genuine auxiliary construction with three copied segments.

2 The Classical Configuration

Let ABC be a nondegenerate isosceles triangle with

$$AB \cong AC.$$

The initial data contain no auxiliary points.

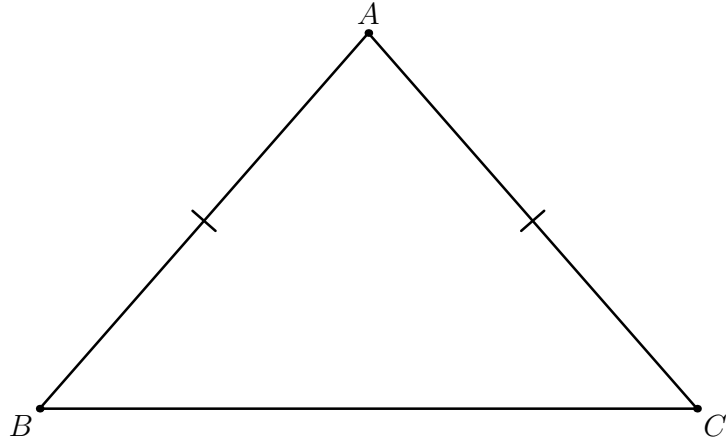


Figure 1: The initial isosceles triangle of Proposition I.5, with $AB \cong AC$.

2.1 Euclid's auxiliary construction

Produce AB beyond B and AC beyond C . The outer points D, E in the formal statement record these produced rays:

$$A - B - D, \quad A - C - E.$$

Euclid then chooses an arbitrary point F on the first produced side. On the other produced side Proposition I.3 cuts off AG so that

$$AF \cong AG.$$

Because $AB \cong AC$, Common Notion 3 gives

$$BF \cong CG.$$

Finally join F to C and G to B .

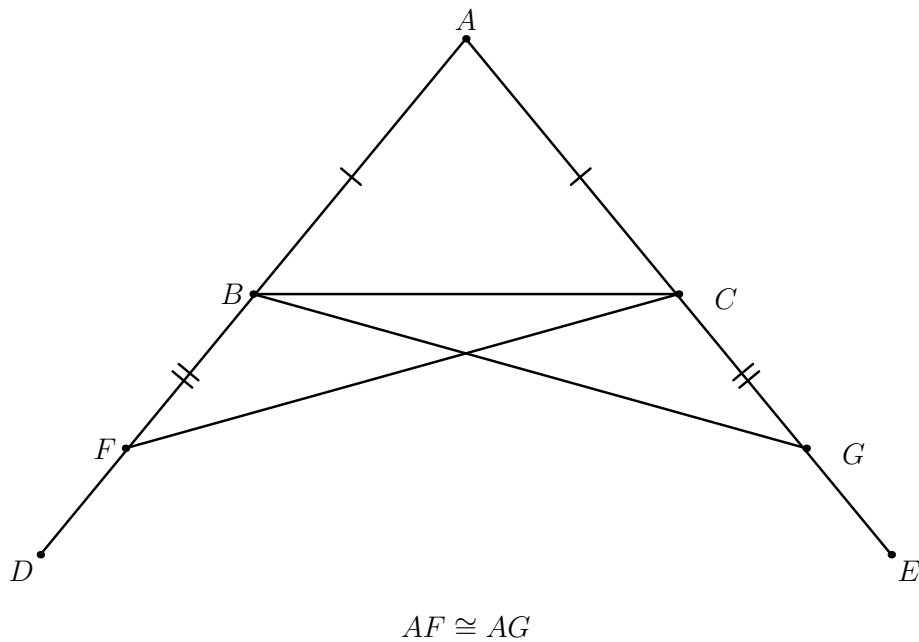


Figure 2: The complete classical configuration. The direct I.3 construction is $AF \cong AG$; after subtracting the given $AB \cong AC$, Common Notion 3 gives $BF \cong CG$. The cross-joins FC and GB prepare the two SAS comparisons.

This corrects an ambiguity in the preserved chapter: G is not introduced by directly prescribing $CG \cong BF$. Euclid prescribes $AG \cong AF$; the shorter equality follows by subtraction.

3 Statement

Theorem 1 (Euclid I.5). *Let ABC be a nondegenerate triangle satisfying*

$$AB \cong AC.$$

Let the equal sides be produced beyond the base vertices so that

$$A - B - D, \quad A - C - E.$$

Then

$$\angle ABC \cong \angle ACB$$

and

$$\angle CBD \cong \angle BCE.$$

The production Lean declaration has the corresponding pair-valued conclusion:

```
theorem euclid_proposition_5
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E : Geo.Point)
  (hABC : not Collinear Geo A B C)
  (hABAC : Geo.Congruent A B A C)
  (hABD : Geo.Between A B D)
  (hACE : Geo.Between A C E) :
  Geo.AngleCongruent A B C A C B /\
  Geo.AngleCongruent C B D B C E := by
  ...
```

Diagrammatic audit. The preserved Book1R chapter had already made one good compression: the successive classical snapshots were merged into one complete auxiliary configuration. That decision is retained. The input triangle is also kept, because it separates the theorem data from Euclid’s larger reductio-free construction.

The old formal architecture flowchart is replaced by geometry. The formal figure has two panels. Panel (a) displays the Pappus-style SAS comparison hidden in `hilbert_isosceles_base_angles`: the same geometric triangle is read as ABC and ACB . Panel (b) records the genuinely new construction inside Hilbert Theorem 14: three segments from the second linear pair are copied onto the three rays of the first angle, producing witnesses X, Y, Z .

Thus the audited set contains three figures and four panels.

4 Classical Proof

The first SAS comparison is between the triangles

$$\triangle FAC \quad \text{and} \quad \triangle GAB.$$

The two side pairs are

$$FA \cong GA, \quad AC \cong AB,$$

and the included angles at A are the same angle formed by the two produced sides, with the rays read in the opposite order. Proposition I.4 therefore yields, among other consequences,

$$FC \cong GB$$

and the corresponding angle congruences needed for the second comparison.

Now compare

$$\triangle FBC \quad \text{and} \quad \triangle GCB.$$

The side equalities are

$$FB \cong GC, \quad FC \cong GB,$$

and the included angles at F, G are congruent by the first SAS result and same-ray identification along the produced sides. A second application of I.4 gives the equality of the angles under the base. Since F and D lie on the same produced ray from B , and G and E lie on the same produced ray from C , this is precisely

$$\angle CBD \cong \angle BCE.$$

The first SAS comparison also gives equality of two larger angles containing the original base angles. The second SAS comparison gives equality of the parts cut off from them. Common Notion 3 therefore subtracts equal angles from equal angles and yields

$$\angle ABC \cong \angle ACB.$$

The direct classical dependency chain is therefore

$$\text{I.3} + 2 \times \text{I.4} + \text{Common Notion 3} \longrightarrow \text{I.5}.$$

Proposition I.2 is an indirect dependency through I.3, not an additional proposition-level step in Euclid's I.5 proof.

5 Proof Architecture of the Reconstruction

At the public Book I level the proof is exceptionally compact:

$$\begin{array}{c}
 AB \cong AC \\
 \downarrow \\
 \text{hilbert_isosceles_base_angles} \\
 \downarrow \\
 \angle ABC \cong \angle ACB \\
 \downarrow \quad A - B - D, \quad A - C - E \\
 \text{hilbert_adjacent_angles_congruent} \\
 \downarrow \\
 \angle CBD \cong \angle BCE.
 \end{array}$$

The source contains exactly these two substantive calls:

```

have hBase :
  Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
  Geo A B C hABC hABAC

have hExterior :
  Geo.AngleCongruent C B D B C E :=
  hilbert_adjacent_angles_congruent
  Geo
  A B C D
  A C B E
  hABD
  hACE
  hABC
  hACB
  hBase

```

The only additional local step is the rotated noncollinearity hypothesis A, C, B required by the second theorem.

6 The Two Hidden Formal Engines

6.1 The isosceles theorem: Pappus-style SAS

The theorem `hilbert_isosceles_base_angles` contains no auxiliary construction. It compares the same triangle in the two vertex orders

$$\triangle ABC \quad \text{and} \quad \triangle ACB.$$

The side data are $AB \cong AC$ and its symmetric form $AC \cong AB$. The included angle at A is congruent to itself after reversing its two rays. A single call to SAS therefore returns the base-angle congruence.

This is the classical Pappus shortcut in formal form. It is mathematically substantive but creates no new point, line, ray, or incidence relation.

6.2 Hilbert Theorem 14: adjacent congruent angles

The second theorem is deeper. In its general form it begins with two linear pairs

$$A - O - C, \quad A' - O' - C',$$

and a congruence

$$\angle AOB \cong \angle A'O'B'.$$

It constructs three points on the rays of the first angle:

$$X \in \text{ray}(OA), \quad Y \in \text{ray}(OB), \quad Z \in \text{ray}(OC),$$

with

$$OX \cong O'A', \quad OY \cong O'B', \quad OZ \cong O'C'.$$

The proof then uses the previously derived SAS consequences, transports the betweenness $A - O - C$ to $X - O - Z$, applies segment additivity, performs a second SAS comparison, and finishes with a final SAS in the adjacent orientation. This is Hilbert's neutral Theorem 14.

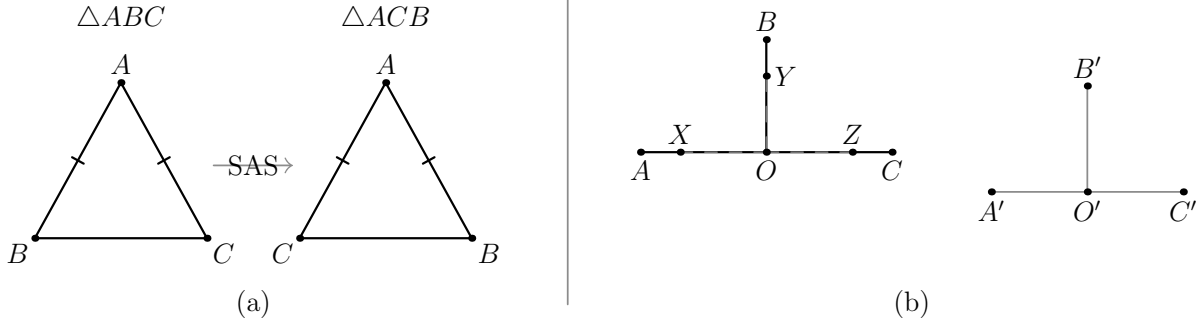


Figure 3: Formal engines below Proposition I.5. (a) The base-angle theorem compares the same isosceles triangle as ABC and ACB and closes by SAS. (b) In the general adjacent-angle theorem, $A - O - C$ and $A' - O' - C'$ are linear pairs; the proof copies $O'A'$, $O'B'$, $O'C'$ onto the rays OA , OB , OC , producing X, Y, Z before the SAS/additivity chain.

When Theorem 14 is specialized to Proposition I.5, the first linear pair is $A - B - D$ and the second is $A - C - E$. The public theorem therefore inherits all of this neutral construction even though none of the witnesses appears in the statement of `euclid_proposition_5`.

7 Hilbert Reconstruction

The formal reconstruction does not replay Euclid's points F, G or his two applications of Proposition I.4. Instead the two classical outputs have already been separated into reusable Hilbert theorems.

For the interior base angles, triangle symmetry plus SAS is enough. For the angles below the base, the two betweenness hypotheses provide the linear pairs required by Hilbert Theorem 14. Thus the formal architecture is modular in a precise sense: the Book I theorem packages two independently proved neutral results rather than hiding a new proposition-local construction.

No Euclidean parallel axiom occurs in either formal engine. The proof belongs to the incidence/order/congruence layer.

8 Lean Formalization

The production theorem first obtains

```
have hBase :
  Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
  Geo A B C hABC hABAC
```

and rotates noncollinearity:

```
have hACB : not Collinear Geo A C B := by
  intro h
  exact hABC (PrimCollinearRotate Geo A C B h)
```

It then invokes Hilbert Theorem 14 through `hilbert_adjacent_angles_congruent` and returns

```
exact And.intro hBase hExterior
```

The current `Proposition05.lean` contains no proposition-local helper theorem, no local axiom declaration, and no geometric case split. This documentation audit read the production source on the repository default branch; it did not run a fresh `lake build` or `#print axioms` check, so no stronger compilation claim is made here.

9 Relationship with Earlier Propositions

9.1 Proposition I.3

I.3 is the direct constructional dependency in Euclid’s proof. After F is chosen on one produced side, I.3 cuts from the other produced side the segment AG satisfying $AG \cong AF$.

The earlier chapter wording that described G as if $CG \cong BF$ were the prescribed construction conflated the construction with its first derived consequence. The equality $BF \cong CG$ follows only after subtracting the given $AB \cong AC$ from $AF \cong AG$.

9.2 Proposition I.4

I.4 is used twice classically. The first SAS comparison creates the cross side equality $FC \cong GB$ and the auxiliary angle congruences. The second uses those results together with $BF \cong CG$ to obtain the angles below the base and the equal angle-parts needed for the final subtraction.

Formally the SAS principle appears lower in the architecture. It is the entire proof engine of `hilbert_isosceles_base_angles`, and it also appears inside the independent proof of Hilbert Theorem 14.

9.3 Transition to Proposition I.6

I.5 proves the forward implication

$$AB \cong AC \implies \angle ABC \cong \angle ACB.$$

Proposition I.6 proves the converse. In the present Hilbert architecture the two Book I propositions are therefore short endpoints of different general congruence results rather than mutually dependent formal proofs.

10 Hilbert Components Used

10.1 `hilbert_isosceles_base_angles`

This is Hilbert Theorem 11 in the current interface. Its proof is the Pappus-style SAS comparison of ABC with ACB .

10.2 hilbert_adjacent_angles_congruent

This is Hilbert Theorem 14. Its source comment records the mathematical engine explicitly: three points are constructed on the rays of the first angle; Theorem 12 is used twice, segment additivity III.3 once, and the final rigidity step is SAS III.5.

10.3 Order and same-ray transport

The relations $A - B - D$ and $A - C - E$ are not decorative. They provide the two linear pairs required by Theorem 14 and allow angle representatives to be replaced along the corresponding rays without changing the angles.

11 Dependency Summary

The classical and formal DAGs are genuinely different.

Classical Euclid

```
AB ~= AC
|
produce AB and AC
|
choose F; I.3 constructs G with AF ~= AG
|
CN3: BF ~= CG
|
I.4 on FAC and GAB
|
FC ~= GB + angle data
|
I.4 on FBC and GCB
|
angles below base equal
+ CN3 subtraction of equal angles
|
base angles equal
```

Formal reconstruction

```
AB ~= AC
|
hilbert_isosceles_base_angles
(Pappus-style SAS)
|
angle ABC ~= angle ACB
|
A-B-D, A-C-E
|
hilbert_adjacent_angles_congruent
```

(Hilbert Theorem 14)

|
angle CBD ~= angle BCE

No coordinate representation, numerical length, numerical angle measure, or Euclidean parallel principle is introduced.

12 Conclusion

Proposition I.5 is an instructive boundary between the historical and formal architectures of Book I. Euclid builds a sizeable auxiliary configuration so that Proposition I.4 can be used twice. The current formal theorem is short because both outputs have already been abstracted into neutral Hilbert infrastructure.

The audit exposes both sides of that fact. The classical figure retains the full F, G construction and corrects the distinction between the prescribed $AF \cong AG$ and the derived $BF \cong CG$. The formal figure removes the old flowchart and shows the actual mathematics below the two interface calls: a Pappus-style SAS proof for the base angles and Hilbert's three-witness construction for adjacent angles.